

CPH 200C PS3 · CLINICAL AI BY REGION

Three Operating Systems for Clinical AI

Why deployed clinical AI looks different across the US, Europe, and Asia: policy, funding, technical developments, and landscape

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CPH 200C · Computational Precision Health

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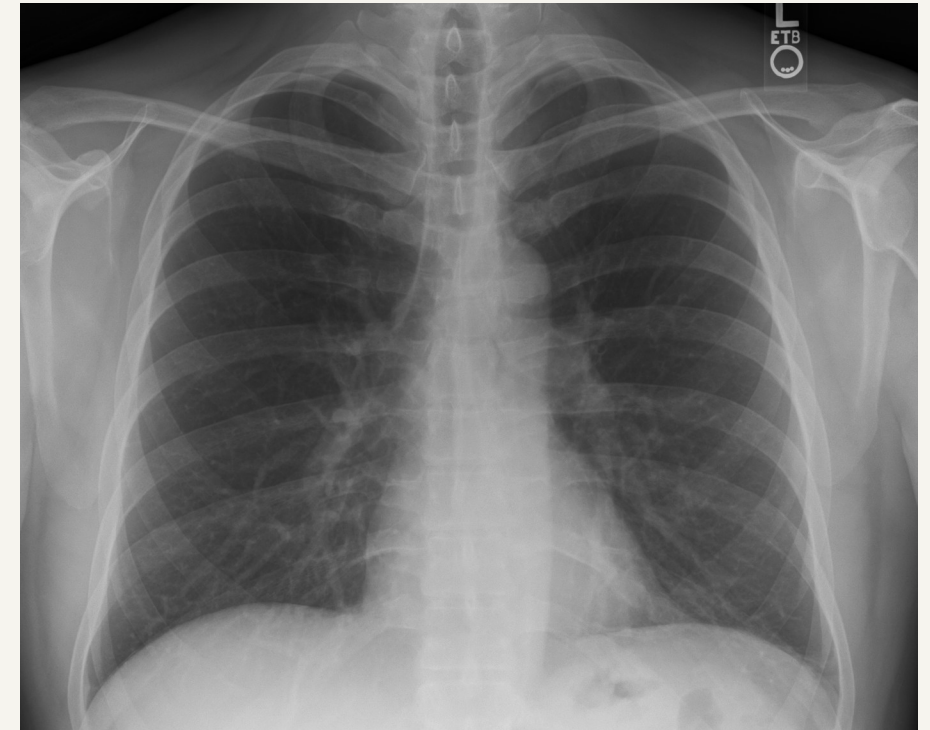
The lens: deployed clinical AI

Three regions, three operating systems for AI

UNITED STATES	EUROPE / EU	ASIA
Market-led	Comprehensive	China-led
Light federal + state patchwork; deregulate to 'win the race'. Closed frontier labs, deepest capital.	Horizontal risk-tiered AI Act (now softening). Sovereignty money, little frontier building.	State-driven core (China: open-weight, population scale). Korea/Japan/India build their own models. Asia owns the AI hardware: TSMC, HBM, Foxconn.

Three operating systems (US / EU / Asia), tested on one case.

Evidence base: npj Digital Medicine - Nature Medicine - Lancet Digital Health - JAMA
- JMIR - Stanford AI Index 2026 - MIT NANDA



Deployed clinical AI is mostly radiology, the richest-evidence test case.

Chest radiograph: public domain, Wikimedia Commons.

What leaders predicted, and what's different now

CHINA

Kai-Fu Lee

AI Superpowers (2018)

The age of implementation: execution and data, not breakthroughs, decide the race, favoring China.

UNITED STATES

Eric Schmidt et al.

The Age of AI (2021); NSCAI

AI is a national-security race the US must win; warned of China closing the gap fast.

EUROPE / EU

Mario Draghi

EU Competitiveness Report (2024)

Europe regulates but does not build; structurally behind on compute, capital, and scale.

WHAT'S DIFFERENT NOW (2025):

None predicted a Chinese OPEN model (DeepSeek) would reset the frontier while US labs re-closed. Lee's implementation thesis holds, but the ROI gap shows capability is not yet value.



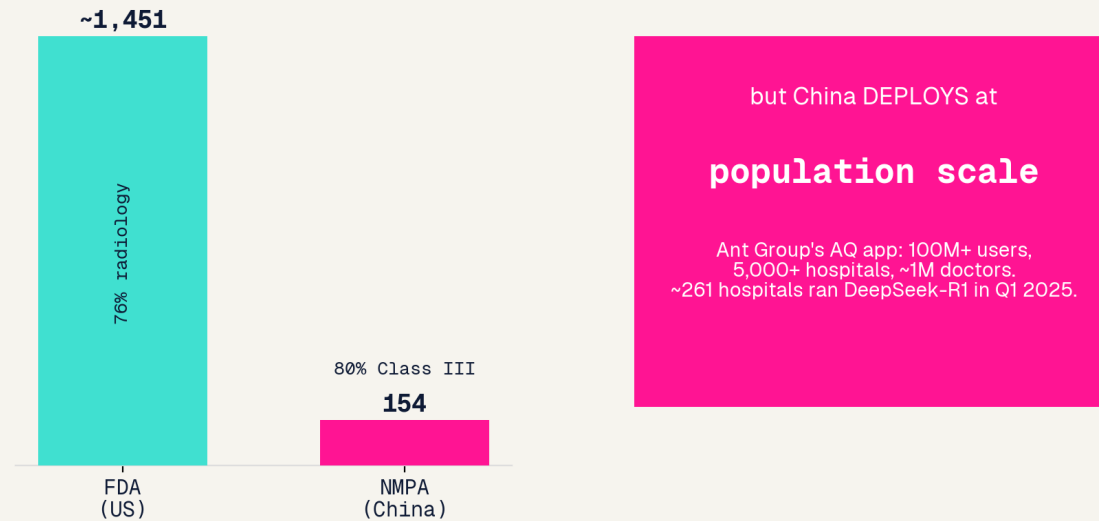
POLICY

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Regulating clinical AI

Clinical AI: the US approves the most discrete tools; China deploys at the most scale

Cumulative AI device approvals



Han et al. (Lancet Digit Health 2024): most clinical-AI RCTs remain few and early-stage.

United States

Device-by-device 510(k) + PCCP. ~1,451 AI/ML devices cleared, 76% radiology; generative/LLM CDS still has no pathway.

[Bedi et al., npj Digit Med 2025](#)

Europe / EU

AI Act high-risk stacked on MDR/IVDR (double conformity); few dual-designated notified bodies; health-data space deferred to 2029.

[EU AI Act + MDR/IVDR](#)

Asia (China)

State-coordinated NMPA: 154 AI devices, ~80% the higher-risk Class III with mandatory clinical trials.

[China NMPA AI-device review, JMIR 2026](#)



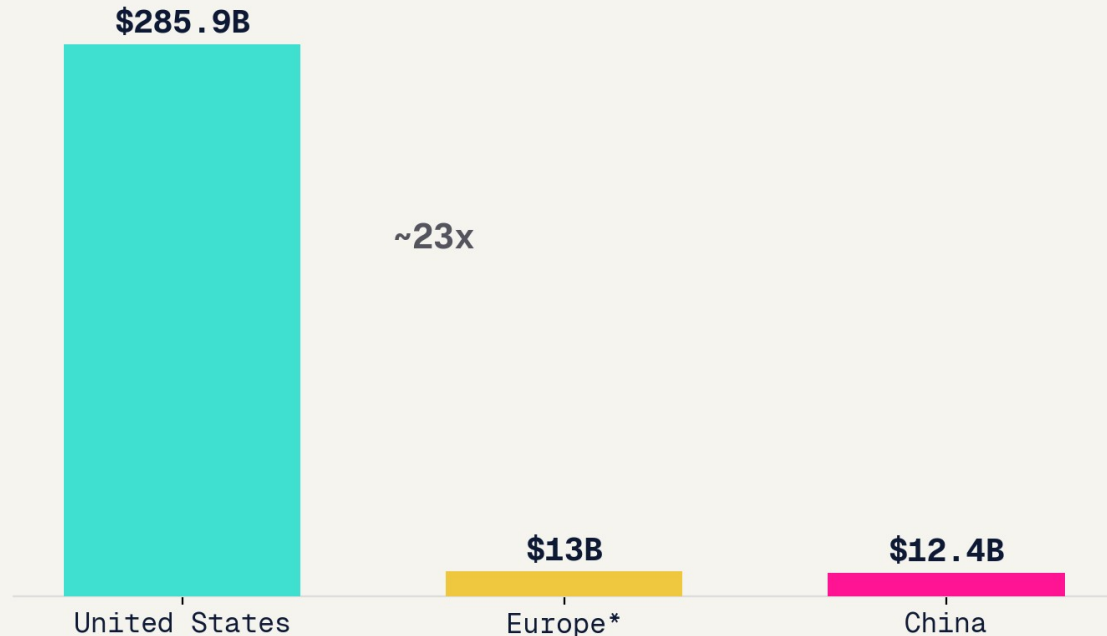
FUNDING

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Capital versus reimbursement

Private AI investment, 2025

Stanford AI Index (US, China). *Europe = est. EU+UK AI VC. The US captured ~83% of global private AI investment.



Stanford AI Index 2026. US ~83% of global private AI investment.

United States

Capital-rich, reimbursement-poor: mostly temporary CPT Category III codes + capped NTAP; few permanent Medicare pathways.

Health Tech Investment Act 2025

Europe / EU

National fast-tracks. Germany's DiGA reimburses ~60 apps (>1M prescriptions); France's PECAN copied it.

DiGA / BfArM

Asia (China)

State funding + volume-based procurement drive deployment without a Western-style reimbursement market.

NMPA / state procurement

Insurers also use AI to deny coverage.
Mello & Rose, JAMA Health Forum 2024

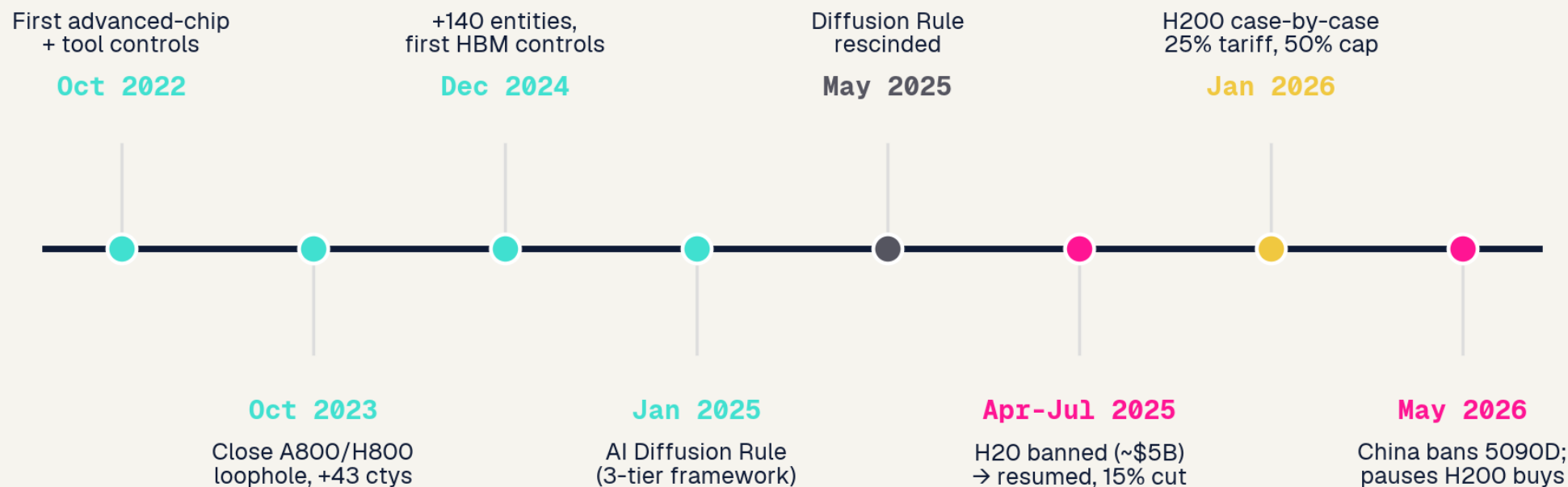


TECHNICAL DEVELOPMENTS

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The chip war reshapes the race

The chip war, 2022-2026: escalation, reversal, and China's pivot to domestic silicon



From the Oct 2022 controls to the Jan 2026 H200 case-by-case reversal, the GPU ceiling is the constraint that shapes every region's technical strategy.

Scale versus constraint: two paths to better models

UNITED STATES: SCALE

Answer the race with compute: Stargate (\$500B), ~\$700B hyperscaler capex in 2026, nuclear power deals. Frontier models stay closed. Abundant GPUs mean less pressure to economize.

Stargate; hyperscaler capex 2026

CHINA: CONSTRAINT

Capped on advanced GPUs by export controls, efficiency becomes the edge: mixture-of-experts, distillation, DeepSeek's ~\$6M training run, cheap inference. Necessity drives method innovation.

DeepSeek-V3 technical report, 2025

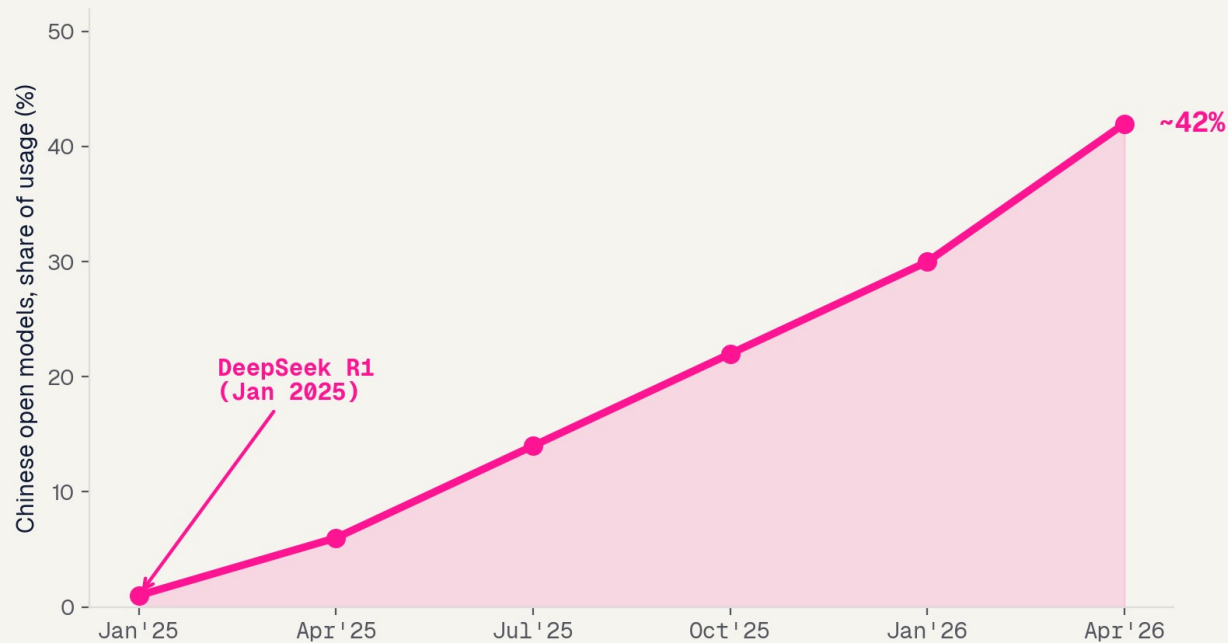
EUROPE: the upstream chokepoint. ASML's EUV monopoly is the gate the controls run through.

The provocative read: the GPU ceiling may be pushing China toward more genuine methodology innovation, while the US mostly scales hardware.

Open weights reach the bedside

China's open-weight surge

Open models (DeepSeek, Qwen) rose from ~1% to ~40%+ of OpenRouter usage in a year; HF downloads China 17.1% > US 15.9%



Chinese open models, share of usage on a major model router.

The DeepSeek moment

An open, frontier-quality reasoning model at low cost reset the board in Jan 2025; ~\$600B wiped off NVIDIA in a day.

NVIDIA -17% selloff, Jan 2025

Into hospitals, fast

~261 Chinese hospitals locally deployed DeepSeek-R1 in Q1 2025 (84% tertiary), outpacing regulation.

DeepSeek-R1 in 261 hospitals, Nat Med 2025

The Western void

FDA's Jan 2026 CDS guidance is silent on generative/LLM clinical AI; the EU defaults it to high-risk.

FDA CDS guidance, Jan 2026

RCT caution: an LLM did not significantly improve physician diagnosis. Goh et al., JAMA Netw Open 2024

Evaluating clinical AI: can we even measure it?

HEALTHBENCH

Physician-built rubrics (262 doctors, 5,000 conversations) grade how well LLMs answer real health questions, the first serious clinical-LLM benchmark.

Arora et al., HealthBench, 2025

HEALTHADMINBENCH

Tests computer-use agents on real healthcare administration tasks, prior auth, billing, scheduling, the unglamorous work that dominates cost.

Bedi et al., HealthAdminBench, 2026

The limit: benchmarks measure what a model can say, graded against expert opinion, not whether it improves patient outcomes. We are getting good at testing capability, not benefit.



LANDSCAPE

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Who deploys clinical AI at scale

UNITED STATES

- ~1,451 AI/ML devices cleared; 76% radiology
- Reimbursement-gated: temporary CPT III codes
- Slowest path from clearance to bedside

Bedi et al., npj Digit Med 2025

EUROPE / EU

- AI Act high-risk + MDR/IVDR double burden
- Nationwide AI mammography deployed (Germany)
- DiGA the working reimbursement route

Eisemann et al., Nat Med 2025

ASIA (CHINA-LED)

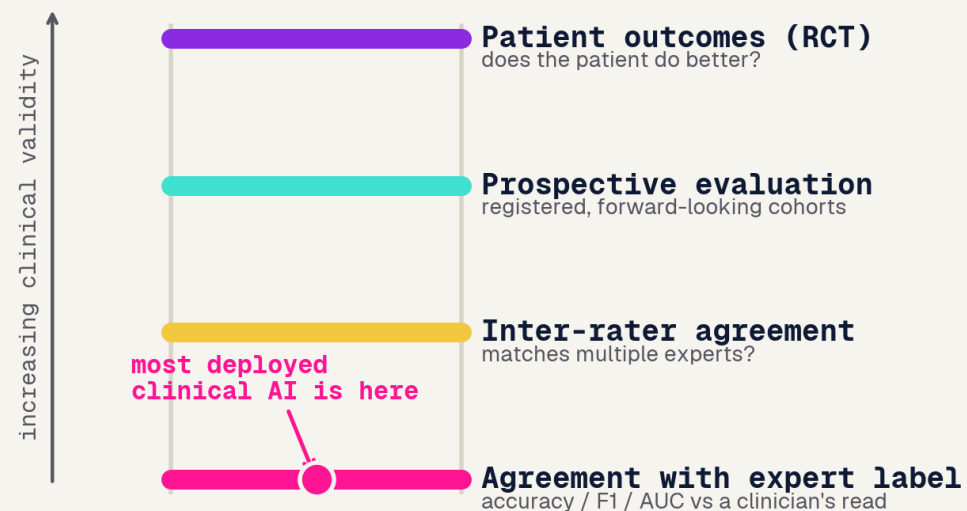
- Ant Group AQ app: 100M+ users, 5,000+ hospitals
- ~261 hospitals ran DeepSeek-R1 in Q1 2025
- Korea exports: Lunit FDA-cleared, ~1M mammograms/yr

DeepSeek-R1 in 261 hospitals, Nat Med 2025

Approvals and deployment are different axes: the US leads on count, Asia on scale.

The ROI reality

The validation gap: agreement is not benefit



Agreement with a label is not benefit to a patient.

95%

of enterprise gen-AI pilots showed no measurable P&L impact (MIT NANDA 2025)

Who closes the validation gap?

UNITED STATES

Reimbursement scrutiny forces some evidence, but approval is still label-based.

EUROPE / EU

Demands evidence ex-ante (AI Act + MDR), yet the health-data space is deferred to 2029.

ASIA (CHINA)

Deploys before outcomes are proven, the fastest rollout, the widest validation gap.

Han, Lancet Digit Health 2024 - Challa et al. (MIT NANDA) 2025

Clinician and AI: help or hazard?

THE PROMISE

AI augments the clinician, extending scarce expertise and catching what a tired human misses. The pitch behind every deployment.

THE RCT REALITY

Giving physicians an LLM did not significantly improve diagnostic reasoning vs usual resources, the model alone scored well; doctors underused it.

Goh et al., JAMA Netw Open 2024

THE HAZARD

Over-reliance breeds automation bias and deskilling, e.g. endoscopists detected fewer adenomas after routine AI use.

Budzyn et al., Lancet Gastro 2025

Success hinges on the human-AI interaction, not accuracy. And the evidence is lopsided: the US and EU run these RCTs and automation-bias studies; China's population-scale deployment has outpaced interaction research.

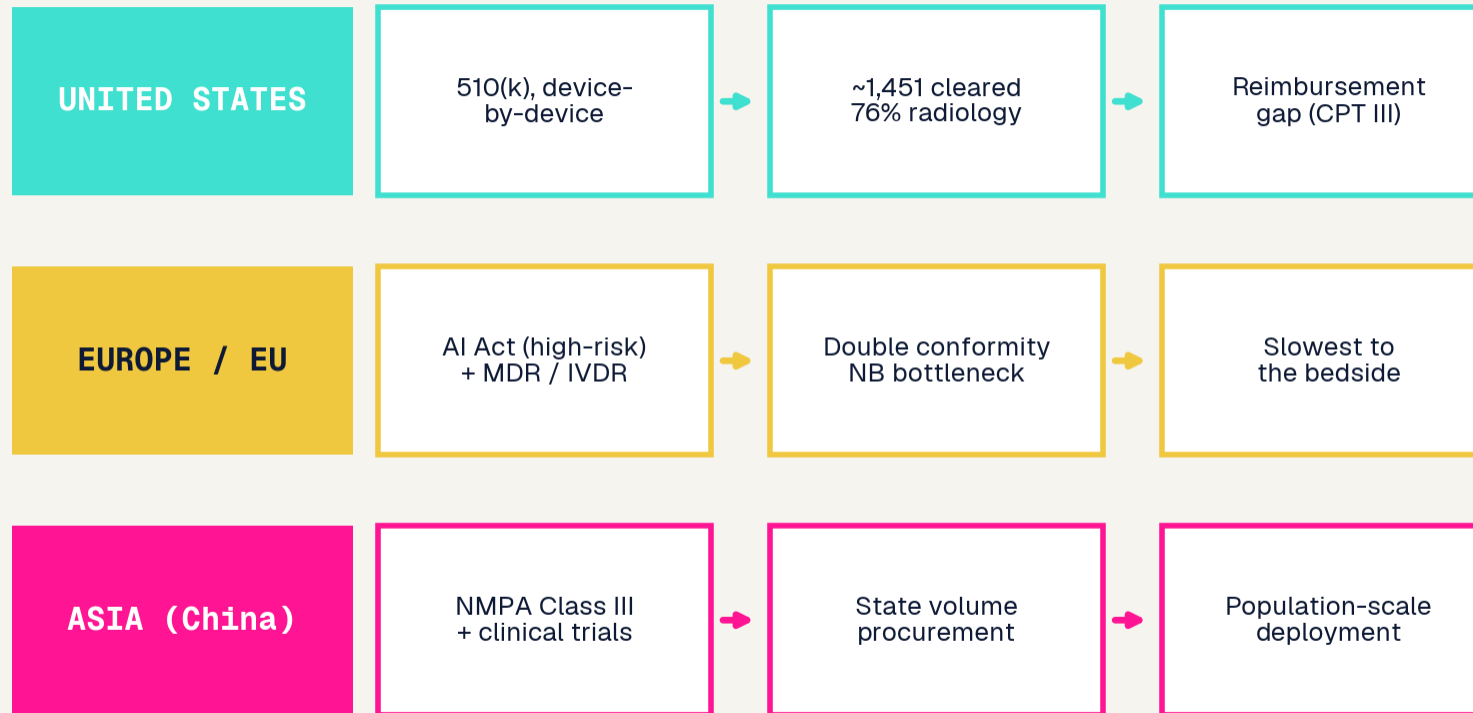


SYNTHESIS

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Regulate, build, deploy, in medicine

How clinical AI reaches patients: three pathways



The bottleneck on deployed clinical AI is institutional, regulation, payment, and coordination, not model quality.

Takeaways and outlook

01

The bottleneck on deployed clinical AI is institutional, regulation, payment, coordination, not model quality.

02

Same three operating systems define clinical AI: the US approves, Europe regulates, Asia deploys.

03

Open weights reach the bedside fastest under state coordination; the US and EU still lack a generative-clinical-AI pathway.

04

Measure what matters: shift clinical-AI evaluation from agreement-with-labels to patient outcomes.

Key references

#	Reference	Venue / year
1	Han et al. - RCTs evaluating AI in clinical practice	Lancet Digital Health 2024
2	Goh et al. - LLM influence on diagnostic reasoning (RCT)	JAMA Network Open 2024
3	Eisemann et al. - Nationwide AI in mammography screening	Nature Medicine 2025
4	Bedi et al. - FDA AI/ML-enabled device landscape	npj Digital Medicine 2025
5	Chen et al. - AI medical devices approved by China NMPA	JMIR Med Informatics 2026
6	DeepSeek-R1 deployed across Chinese hospitals	Nature Medicine 2025
7	Mello & Rose - AI tools and insurance coverage denials	JAMA Health Forum 2024
8	Stanford HAI - AI Index Report 2026	Stanford HAI 2026
9	Challa et al. (MIT NANDA) - State of AI in Business	MIT NANDA 2025
10	Arora et al. - HealthBench	arXiv 2025

Framing: Lee, AI Superpowers (2018) - Schmidt et al., The Age of AI (2021) - Draghi, EU Competitiveness Report (2024).

Full source list with URLs in the companion research.md.

Thank You

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夢想